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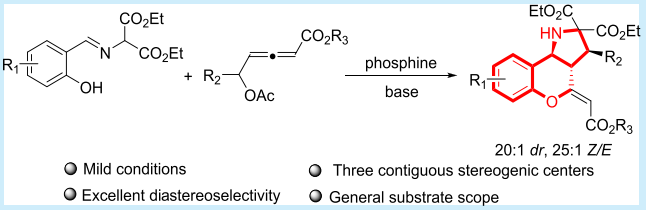
Phosphine-Catalyzed Stereoselective Tandem Annulation Reaction for the Synthesis of Chromeno[4,3-*b*]pyrroles

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ABSTRACT: A phosphine-catalyzed tandem cyclization reaction has been developed to provide a series of chromeno[4,3-*b*]pyrrole derivatives, which contain three consecutive asymmetric centers. The reaction has a good yield, excellent stereoselectivity, and *Z*/*E* selectivity. The new method is simple, requires only mild conditions, and shows tolerance for various functional groups. Similarly, this reaction can be catalyzed by a chiral phosphine catalyst to achieve asymmetric synthesis.



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The chromenopyrrole derivatives are ubiquitous structural options founded in numerous natural and biologically active compounds (Figure 1).[1,2] For instance, compound **a** is

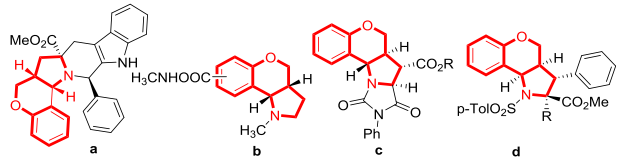
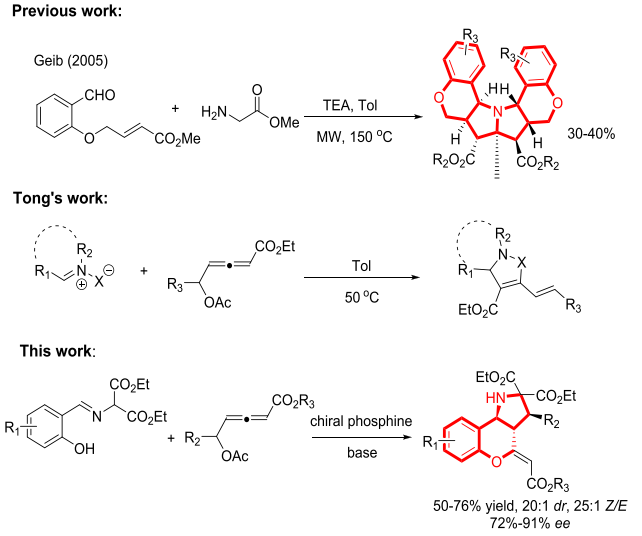


Figure 1. Representative bioactive chromeno[4,3-*b*]pyrrole structures.

an antagonist of muscarinic acetylcholine receptors (mAChRs) and displays potent activity against four selected bacterial pathogens.[2a] Product **b** forms a lead for the development of new AChE inhibitors.[2b] In addition, chromeno[4,3-*b*]pyrrole derivatives were reported to possess anticancer, antimicrobial, and antioxidant activity.[2c−f] Therefore, the development of a facile and stereoselective methodology for constructing chromeno[4,3-*b*]pyrrole frameworks has been a long-standing goal in organic and medicinal chemistry.

To date, numerous methodologies for obtaining the chromenopyrrole have been reported because of their remarkable biological activities.[3] For example, in 2005, Geib's group successfully pioneered a double intramolecular cyclization of *O*-allylic salicylaldehydes to synthesize chromeno[4,3-*b*]pyrroles (Scheme 1). However, some of these protocols suffered from some undesired limitations such as excessive temperatures, toxic solvents, or complex experimental operations. Considering the huge potential application value of these compounds in drug discovery, exploiting expeditious and efficient methodologies to access structurally diverse and complex chromeno[4,3-*b*]pyrroles is highly desirable. Since the report of Lu's [3+2] cycloaddition, a great deal of research has been devoted to exploring the

Scheme 1. Methods of Synthesizing Chromeno[4,3-*b*]pyrrole and a 1,3-Dipole [3+2] Cycloaddition Strategy



cycloaddition reaction catalyzed by organophosphines.[4] These methods have become sturdy synthetic tools for the construction of a wide range of carbo- and heterocycles.[5] In terms of mechanism, phosphine-catalyzed annulations can be carried out by zwitterions, which were directly formed by 1,4-addition of a phosphine catalyst to activated allenes. These zwitterions exhibit excellent dipolar carbon synthon properties

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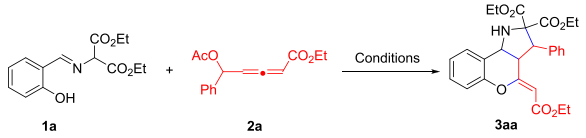
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and can react with diverse electrophiles.[6] In 2015, Tong's group reported that as a versatile dipolarophile, δ-acetoxy allenoates had the ability to undergo 1,3-dipolar cycloaddition reactions with various dipoles.[7a] In the presence of a phosphine catalyst, the allenoates can be turned into 2-phosphonium diene intermediates, which can react with an array of bisnucleophiles.[7] Herein, we report a phosphine-catalyzed cascade cycloaddition reaction between aldimine esters and δ-acetoxy allenoates, which delivered diastereoselective chromenopyrrole derivatives.

Our investigation began with the screening of the catalysts for the reaction of aldimine ester **1a** and allenoate **2a** in toluene at room temperature (Table 1, entries 1−4). Weakly

Table 1. Optimization of the Reaction Conditions[a]

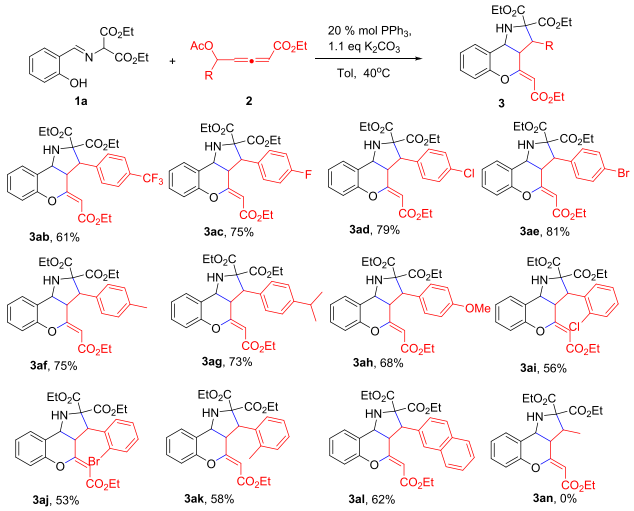


entry	catalyst	solvent	temp[b]	yield[c] (%)
1	PPh_3 (20 mol %)	PhMe	rt	55
2	$PEtPh_2$ (20 mol %)	PhMe	rt	26
3	PEt_2Ph (20 mol %)	PhMe	rt	<10
4	PBu_3 (20 mol %)	PhMe	rt	trace
5	PPh_3 (20 mol %)	THF	rt	31
6	PPh_3 (20 mol %)	$CHCl_3$	rt	43
7	PPh_3 (20 mol %)	CH_2Cl_2	rt	20
8	PPh_3 (20 mol %)	CH_3CN	rt	27
9	PPh_3 (20 mol %)	Et_2O	rt	<10
10	PPh_3 (20 mol %)	acetone	rt	trace
11	PPh_3 (20 mol %)	PhMe	40 °C	83
12	PPh_3 (20 mol %)	PhMe	60 °C	62
13	PPh_3 (20 mol %)	PhMe	80 °C	42

[a]To the solution of **2a** (31 mg, 0.12 mmol, 1.2 equiv), catalyst (0.02 mmol, 0.2 equiv), and K_2CO_3 (0.11 mmol, 1.1 equiv) in solvent (1 mL) was slowly added the solution of **1a** (28 mg, 0.1 mmol, 1.0 equiv) in solvent (1 mL). [b]An oil bath as the heat source. [c]Isolated yield.

nucleophilic phosphines (such as PPh_3) showed good reactivity, and the desired product **3aa** was obtained with a yield of 55%; strongly nucleophilic phosphines (such as PBu_3) could not generate **3aa**. Then the effect of the solvent on the reaction was evaluated (entries 5−10). Compared with toluene, other solvents led to decreased yields or even no target product. Also, temperature has a significant effect on the yield of **3aa** (entries 11−13), and the best result was obtained when the reaction was performed at 40 °C. The structure of **3aa** was undeniably confirmed by X-ray crystallographic analyses.[8] The substrate scope of allenoates **2** was checked for their reaction with substrate **1a** under the optimized reaction conditions as stated in entry 11 of Table 1. As shown in Scheme 2, both electron-withdrawing and -donating substituents could be installed at the *para* position of the phenyl rings of allenoates **2**, with the corresponding chromeno[4,3-*b*]pyrroles (**3ab**−**3ah**) afforded in good yields (61−81%) as single diastereomers. It should be noted that the *ortho* position groups on the benzene ring of allenoates decreased the yield (**3ai**−**3ak**). Furthermore, when a naphthalene ring replaced the benzene ring, the reaction still proceeded normally and gave **3al** in 62% yield. However, for

Scheme 2. Substrate Scope of Allenoates 2[a]

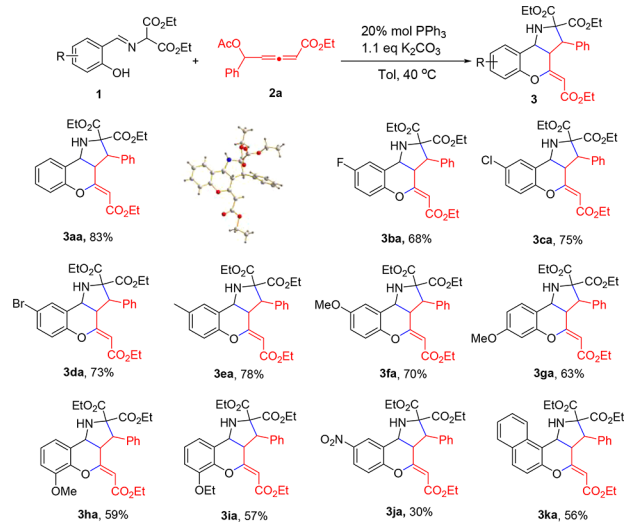


[a]Conditions: 0.1 mmol of **1a**, 0.12 mmol of **2**, 1.1 equiv of K_2CO_3, 20 mol % PPh_3 in 2.0 mL of PhMe at 40 °C (oil bath) for 4 h.

the allenoate with an alkyl substituent, the target compound cannot be obtained (**3an**).

Next, we explored the range of aldimine esters. Generally, all of the aldimine esters tested led to the expected products (Scheme 3). The reactions were tolerant of the electron-

Scheme 3. Substrate Scope of Aldimine Esters 1[a]



[a]Conditions: 0.1 mmol of **1**, 0.12 mmol of **2a**, 1.1 equiv of K_2CO_3, 20 mol % PPh_3 in 2.0 mL of PhMe at 40 °C (oil bath) for 4 h.

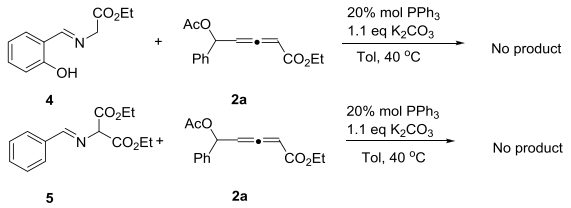
withdrawing groups such as halogens and donating functional groups on the benzene ring. For example, substituents could be fixed at position 4 of the phenol rings in spite of electronic properties, with the corresponding compounds afforded in medium to good yields (**3ba**−**3fa**). The site of the substituents on the benzene ring of the aldimine esters had a weak effect on the yield. As shown in Scheme 3, 5-methoxy-substituted and 6-methoxy-substituted aldimine esters afforded **3ga** in 63% yield and **3ha** in 59% yield. The aldimine ester containing the nitro group made the reaction worse and gave **3ja** in significantly low yield (30%). Furthermore, when the benzene ring was

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replaced with a naphthalene ring, **3ka** was produced in 56% yield.

To recognize these novel processes, ethyl (*E*)-2-[(2-hydroxybenzylidene)amino]acetate **4** and diethyl (*E*)-2-(benzylideneamino)malonate **5** were prepared and treated with **2a** under our standard condition (Scheme 4). However,

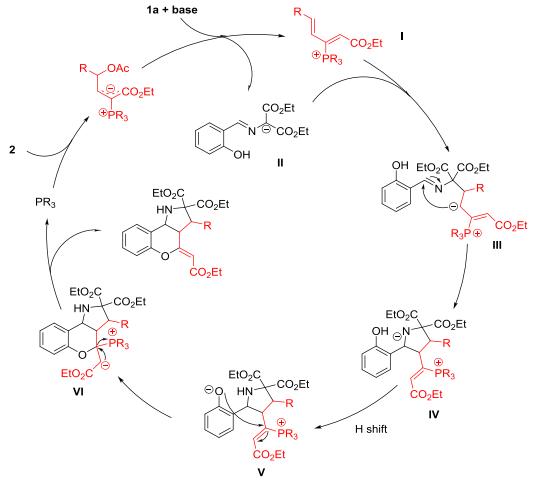
Scheme 4. Control Experiment



no product was obtained. These results illustrated the necessity of the hydroxy group and malonate for the annulation reaction. The hydroxyl at position 2 might activate the malonate and could form intramolecular hydrogen bonds to stabilize the azomethine ylide.[9]

On the basis of the results of our control experiments and previous research, we suggested the possible mechanisms of these domino processes (Scheme 5).[7] Cationic intermediate **I**

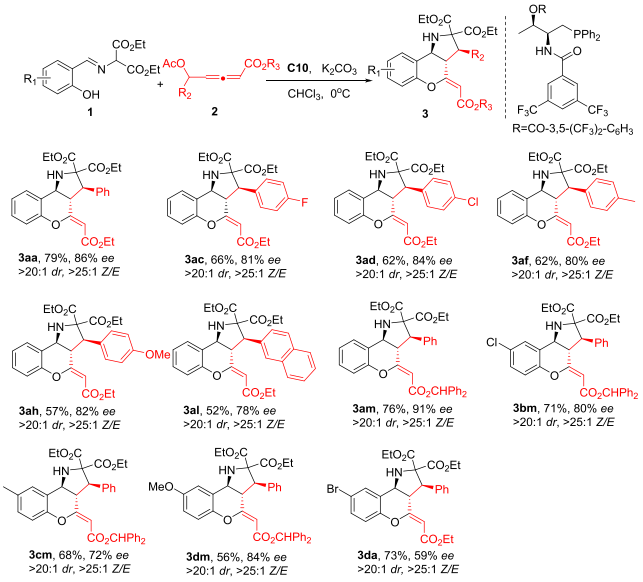
Scheme 5. Proposed Mechanism



was formed by the addition−elimination reaction of phosphine and **2**. Then the carbanion of **II** attacked **I** to generate zwitterionic intermediate **III**, which eventually allowed for intramolecular addition of the carbon anion to vinyl phosphonium to afford intermediate **IV**. After a H-shift process, the resultant intermediate **V** might undergo intramolecular nucleophilic attack, leading to intermediate **VI**. Finally, an elimination yielded product **3** and regenerated the phosphine catalyst.

In addition, we try to explore the asymmetric version of this cycloaddition. Although many chiral phosphine catalysts were investigated, most of them resulted in either a low yield or a low ee value (see the Supporting Information). Only chiral phosphine **C10** gave satisfactory results (Scheme 6). It is particularly worth noting that no matter what phosphine catalysts were used, the reaction has excellent diastereoselectivity and excellent *Z*/*E* selectivity. With **C10** as the catalyst, **3aa** could be obtained in 75% yield with good enantioselectivity (87% ee) when the reaction was performed in $CHCl_3$ at 0 °C. With the aim of further improving the enantioselectivity, we investigated the effect of the substituent of allenoates. Pleasingly, a diphenyl methyl-containing alkene tolerated this reaction, rendering **3am** in 76% yield with 91% ee, 20:1 dr, and 25:1 *Z*/*E* (entry 19). Generally, a variety of allenoates that have different substituents on the benzene ring performed quite well, giving corresponding products **3ac**−**3ah** in moderate yields (57−66%) with good stereoselectivities (all >20:1 dr, >25:1 *Z*/*E*, and 80−84% ee). Furthermore, allonate **2l** could also react with **3a** in the presence of C10, giving **3al** in 52% yield with 78% ee. Subsequently, both electron-withdrawing and -donating substituents on the benzene ring of aldimine esters were found to be tolerant under the optimal conditions, and products **3bm**−**3dm** can be obtained in medium to good yields (56−73%) with good diastereo- and enantioselectivities (all >20:1 dr, >25:1 *Z*/*E*, and 72−84% ee).

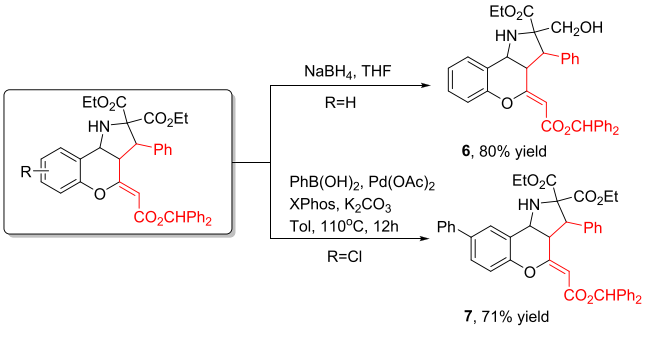
Scheme 6. Preliminary Asymmetric Phosphine-Catalyzed Annulation[a]



[a]Conditions: 0.1 mmol of **1**, 0.12 mmol of **2**, 1.1 equiv of K_2CO_3, 10 mol % **C10** in 2.0 mL of $CHCl_3$ at 0 °C for 12 h.

To highlight the synthetic potential of this method, the transformations of **3** were studied (Scheme 7). First, treatment of **3aa** with $NaBH_4$ in THF afforded the functional chromeno[4,3-*b*]pyrrole **6** in 80% yield. Chromeno[4,3-*b*]pyrrole **3bm** underwent an effective coupling reaction and provided azepine **7** in 71% yield.

Scheme 7. Synthetic Transformations



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In conclusion, we have developed a phosphine-catalyzed cascade cycloaddition strategy for facily synthesizing chromenopyrrole derivatives from aldimine esters and δ-acetoxy allenoates. In particular, a wide range of products have been afforded in generally good yields with excellent diastereoselectivities. In addition, by using a chiral phosphine catalyst, the cycloaddition reaction can also initially realize the asymmetric synthesis of chromeno[4,3-*b*]pyrrole. This protocol can be expected to find wide synthetic application because of its simple operation, mild reaction conditions, high efficiency, and high chemo- and enantioselectivity.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c02558.

Experimental procedures and spectroscopic data for the substrates and products (PDF)

Accession Codes

CCDC 1976344 and 2020535 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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†Z.D. and J.Z. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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